

Implement CI/CD Pipeline Using Jenkins

1. Objective:

- Automate the deployment of Flask and Express applications using Jenkins.

2. Steps:

○ Install Jenkins:

- Install Jenkins on the same EC2 instance or on a separate machine.
- Configure Jenkins by installing essential plugins like Git, NodeJS, and Python.

○ Set Up Jenkins Pipeline:

- Create two separate Jenkins pipelines for the Flask and Express applications.

■ Pipeline Steps:

- Pull the latest code from the respective Git repositories.
- Install dependencies for Flask (`pip install -r requirements.txt`) and Express (`npm install`).

○ Triggering the Pipeline:

- Set up a GitHub webhook to trigger the Jenkins pipeline on every push to the repositories.

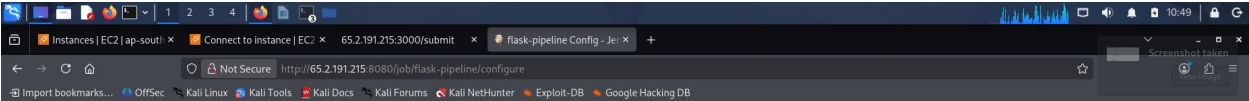
■

○ Optional Enhancements:

- Add testing stages to the pipeline for both applications.
- Configure environment variables in Jenkins for managing sensitive data like API keys.

3. Deliverables:

- A fully functional CI/CD pipeline that automates the deployment process for Flask and Express.
- A Jenkins pipeline script (`Jenkinsfile`) for each application.
- Evidence of the pipeline working (e.g., screenshots of successful builds and deployments).



Configure

☐ This project is parameterized ?

☐ Throttle builds ?

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

☐ Build after other projects are built ?

☐ Build periodically ?

☒ GitHub hook trigger for GITScm polling ?

☐ Poll SCM ?

☐ Trigger builds remotely (e.g., from scripts) ?

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

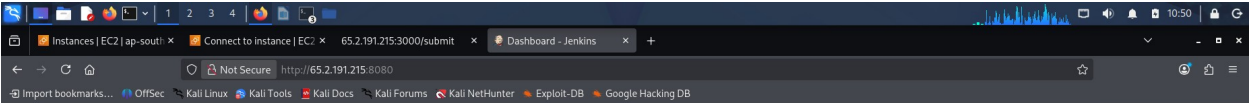
Definition

Pipeline script

Script ?

```
1 pipeline {
```

Save **Apply**



Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

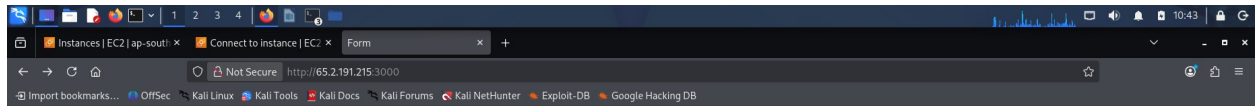
Build Executor Status

0/2

All +

S	W	Name	Last Success	Last Failure	Last Duration
✓	☁	express-pipeline	8 min 9 sec #12	11 min #11	19 sec
✓	☀	flask-pipeline	57 min #2	N/A	5.3 sec

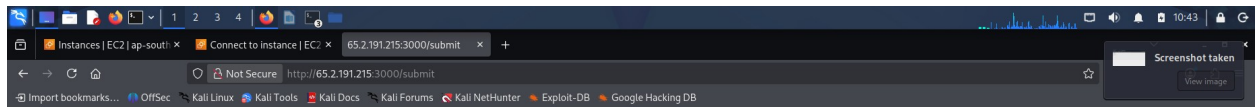
Icon: S M L



User Form

Name:

Email:



Data Received by Flask Backend

Name: Test User
Email: testuser@test.com